Global Facilities - Design & Construction - CSA - Cast-In-Situ Bored Piles Specifications

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1.0 Purpose

1.1 This document provides a Required Standard and Standard Guideline on Material and Design Specification that shall be followed and complied in the Design, Engineering and Construction for Cast-In-Situ Bored Piles as part of any green and/or brown field construction projects contract requirement.

1.2 The standard is applicable for any green and/or brown field construction projects as per required in the contract.

1.3 The document is intended to serve as a reference and guideline in addition to any Construction Drawings and Specification provided in the contract requirement including any required by local authority regulation and standard requirement as well as code of practice.

1.4 The established Specifications Standard for each locale shall also comply with Micron’s in-house:

1.4.1 RBA Code of Conduct requirements for Semiconductor Manufacturing

1.4.2 Environmental, Health & Safety Standards

1.4.3 Operational Requirements /SOP (Standard Operation Procedures)

1.4.4 Security Protocol

1.5 The Specifications Standard to be adopted in each local shall not violate any local codes or by-laws and reflects the current semi-conductor industry’s best practices.

1.6 Where there is the option to adopt either the MasterSpec or in-region specifications standard, the AE (Architectural & Engineering Consultants) shall propose the most suitable solution based on safety, quality, time & cost to Micron's Designated Project Team (DPT) for review and acceptance. Otherwise, the more stringent option shall apply.

2.0 Scope

2.1 This document for cast-in-situ bored piles specification covers all necessary related in design, engineering, procurement, construction, installation and completion of piling to full compliance of the contract requirement and required for the project. It extended to cover any green and/or brown field construction projects.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

3.1 Definitions of acronyms used in this document are listed as follows:

Term/Acronym	Definition / Description
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
AE	External Architectural & Engineering Consultants
QP/PE	External Qualified Person (professional architect/engineer)
DPT	Micron's Designated Project Team
EHS	Micron's Environmental Health & Safety Team
SME	Micron's Subject Matter Expert
TM	Micron's Team Members

4.0 General

4.1 Design

All materials and work shall be in accordance with drawings, this specification and all current Codes of Practice, Euro codes, SS, or equivalent local/international code requirements.

Notwithstanding the above, tests shall be carried out as and when directed by the Engineer in accordance with the relevant standards.

4.2 CODE OF PRACTICES

BS EN 1997-1 Geotechnical design - Part 1: General rules

BS EN 1997-2 Geotechnical design - Part 2: Ground investigation and testing

Or equivalent local/international code requirements in the project location.

5.0 System of Piling

The piling system and equipment proposed by the contractor.

5.1 May include Bucket Auger, Flight Auger, Hydraulic / Rock Augers or Reverse circulation Drill or equivalent system as the principal method for the work in general.

5.2 Shall be capable of sinking holes to the bearing stratum, which is capable of sustaining the working loads with factor of safety as specified.

5.3 Shall be able to deal with whatever strata encountered and ground conditions including negative skin friction, etc.

5.4 Shall be capable of being modified to deal with whatever strata encountered by the addition of further cutting or extraction or protective devices as may be required to deal with:

a. Water-bearing ground

b. Soft ground which caves in unless supported as boring proceeds

c. Firm or stiff clays or marls

d. Uniform Sands

e. Coarse gravels

f. Boulder clays

g. Boulders

h. Thinly bedded shales

i. Tree roots or buried tree trunks

j. Obstruction

5.5 Shall be capable of sinking and removing casings for the support of the boreholes to whatever depth that may be required (to the full depth of the bore if necessary) and without disturbance to the ground surrounding the pile.

5.6 Shall comply with the current Environmental Public Health (Control of Noise from Construction Sites Regulations).

6.0 Warranty

6.1 The Contractor shall on acceptance of his tender, execute a piling warranty to indemnify the Employer from any damage attributable to faulty design, workmanship or materials or to inadequacy of any pile to support the specified working load.

7.0 Technical Information to be Supplied at the Time of Tendering

The tender must be accompanied by details of the system offered together with:

7.1 Method statement of the piling;

7.2 Specification for concrete mix (giving 7 days and 28 days cube strengths) slump allowed, reinforcement and any other materials to be used in the works.

7.3 Method of fabrication and details of length and size of reinforcement cage in each size of pile;

7.4 Method of placing the cage and maintaining in position during concreting;

7.5 Description of method adopted for ensuring a seal at the bottom of casings prior to concreting;

7.6 Details for ensuring the soundness and cleanliness of the pile base prior to concreting;

7.7 Method of concrete placing to prevent segregation;

7.8 Method of safety and safe guards to be adopted, if termite concrete placing system is used for forming the pile.

7.9 Method of forming the belled base if any and ensuring the soundness and cleanliness of the pile base prior to concreting.

7.10 Method of estimating the load carrying capacity in compression for each size of piles proposed to be used assuming that the ground on which the piles to be installed is capable of sustaining a total load of three (3) times the working load before ultimate failure as defined in the specification for ‘Load Test on Piles’ and two (2) times the working load without exceeding the limits of settlements as specified and a group factor to be used for group of piles.

7.11 The basis on which the contractor will decide on the depths of piles to be installed giving the types of test or analysis (field or laboratory) which he proposes to use in determining the bearing capacity of the soil at the bottom of the borehole (i.e. At the bearing stratum).

7.12 Contractor shall undertake to supply to the Engineer any further information that the Engineer may require and deemed necessary.

7.13 Number of piling rigs or frames available and the average output in linear meter of completed or installed piles per normal working day of eight working hours per day.

8.0 Piling To be Carried Out By Competent Persons

8.1 Work shall be carried out by operators and supervisory staff thoroughly experienced in the system. The Contractor shall have available at the site a qualified and experience supervisory staff.

8.2 The Contractor shall submit written evidence of person's experience and qualification.

8.3 The Contractor shall liaise closely with the Engineer at each stage of the work; in particular, the Contractor must first submit his calculations for the proposed carrying capacity of any pile to the Engineer for his approval prior to commencement of piling work.

8.4 All calculations and remedial proposals submit by the contractor shall be endorsed by the Contractor’s Professional Engineer.

9.0 Factor of Safety

9.1 The piles in compression constructed by the Contractor shall be:

a) Capable of carrying the specified working load with a geotechnical factor of safety of 3 and
b) Capable of satisfying the specified acceptance settlement criteria of the standard working load test.

For piles in tension, a minimum factor of safety of 3 against design ultimate tension load is required.

10.0 Installation of Piles – General

10.1 No piling work shall be carried out without the presence of the Engineer or his representative. Checking penetration, nature of soil, measurement of plumb, batter shall be made as and directed by the Engineer.

10.2 Unless otherwise specified or decided by the Engineer, the pile shall be installed (include chiseling) and socketed at least 500mm or half the diameter of pile whichever is the greater into the layer of Hard Strata of RQD exceeding 30%.

10.3 If there is any dispute as to the quality of base founding material, the onus is on the Contractor to carry out coring in the hard strata and determined the rock quality.

The cost of chiseling or rock augering is deemed to be included in the Contractor’s cost.

No claim for variation or time for chiseling or rock augering shall be entertained.

10.4 The onus of obtaining the required resistance to withstand the required test load within the limit of settlement shall rest with the Contractor.

10.5 The Contractor shall cover up and protect the works from weather and suspend all operations which in the opinion of the Engineer will be detrimental to the works.

10.6 The Contractor shall be deemed to have checked the difference between the pile cut-off level and the existing ground level. Any earth filling or building up of piles to the required cut-off level shall be deemed to be inclusive in the tender price. No claims for variation in this respect shall be entertained.

10.7 Existing ground levels at pile positions shall be determined and agreed to between the Contractor and the Engineer before actual piling work starts.

10.8 The installation of all working piles shall not commence until the Contractor has submitted to the Resident Engineer complete calculations showing the load carrying capacity of each pile proposed and its corresponding length of penetration and obtained his approval of such calculations.

10.9 The Engineer reserves the absolute right and the Contractor shall recognize such right to instruct the installation of working piles in any sequence the Resident Engineer deems necessary for the satisfactory completion of the Works. The Contractor will not be allowed to raise any claim for extra payment or compensation on complying with this clause.

10.10 Where five or more piles are to be installed in a group, the center pile shall be installed first with the remaining piles outwards from the center. The Resident Engineer shall have the right and discretion in deciding the period to be allowed for installing adjacent piles in the group of piles.

10.11 No two piles in the same group will be allowed to be constructed on the same day, unless with the approval of the Engineer

10.12 The Contractor shall submit to the Engineer his proposed sequence and timing for boring piles having regard to the avoidance of damage to adjacent piles.

10.13 If during the execution of the work damage is, or is likely to be, caused to existing services or adjacent structures, the Contractor shall submit to the Engineer his proposals for repair or avoidance of such damage.

10.14 The Contractor shall carry out at his own cost, a survey and investigate the presence of existing utility services and buried structures prior to commencement of piling works.

10.15 In any circumstances, the Contractor shall submit design calculations and details of bored piles/micro-piles/pre-cast reinforced concrete piles as alternative proposal for the Engineer’s concurrence prior to commencement of works.

10.16 Such calculations and details shall be prepared and certified by the Contractor’s Professional Engineer, and the Contractor shall make such amendments and changes to the design/details deemed necessary by the Resident Engineer without additional charge.

10.17 The Contractor shall also provide Professional Engineer certified final calculations and details for all the alternate proposal of micro-pile system including capping details etc. for submission to the Building Authority for approval, and shall alter/amend such as calculations/drawings to comply with the requirements of the Authority without additional cost.

11.0 Record of Each Pile

11.1 A record of every pile shall be kept by the Contractor and a copy shall be submitted to the Engineer on site. This record shall include the following information:

(a) Length and diameter of borehole

(b) Ground level

(c) Cut-off level

(d) Length of pile

(e) Length of reinforcement cage

(f) Water table below ground level

(g) Length of casing

(h) Strata of ground penetrated and where boring stops

(i) Result of tests on soils

(j) Concrete cubes

(k) Time of completion.

The form of this record must be approved by the Engineer before piling commences.

12.0 Materials

12.1 Cement

All cement used shall be of an approved brand and manufacture and shall comply in every respect with conditions, analysis tests laid down in the current SS EN 197-1 or local code requirements for Portland Cement.

12.2 Aggregates

The materials used as aggregates shall be chemically inert, strong, hard, durable and free from clay lumps and organic and other impurities. All materials and grading of aggregates shall comply with SS EN 12620, ISO 3310 or local code requirements and tested in accordance with the requirements of SS 73 and BS 812 or local code requirements.

Aggregate used shall not be alkali reactive.

All-in aggregates as specified in SS EN 12620 or local code requirements shall not be used.

Notwithstanding the above, fine aggregates for concrete shall be clean, sharp, well graded sand free from clay, loam or organic matter. Samples shall be submitted by the Contractor for the approval of the Engineer to their use as fine aggregates.

Coarse aggregates for concrete shall be clean crushed granite, well graded between the limits of 20mm and 5mm. Samples shall be submitted by the

Contractor for the approval of the Engineer, prior to their use as coarse aggregates.

12.3 Steel

Steel reinforcement shall be obtained from local manufacturer or other manufacturers as may be approved in writing by the Engineer and shall comply with SS 560 and SS 561 or local code requirements.

All reinforcement shall be free from scale and all loose rust. It shall not be coated with any grease, oil, paint or adhering earth that may impair the bond between the concrete and reinforcement.

13.0 Ready-Mix Concrete

13.1 Where Ready-Mix Concrete is used, it shall comply with SS EN 206-1, SS 544-1, SS 544-2 or local code requirements and the mix design and admixture used shall be approved by the Engineer.

13.2 The ready-mix concrete shall comply in all respects with the specified requirements for concrete as outlined in this Specification, and the Contractor shall arrange and pay for the making of the entire requisite test.

13.3 Ready-mixed concrete shall only be transported in continuous mixing trucks comply with the manufacturer’s operating standard. Appropriate retarder when used shall be subject to the approval of the Engineer.

13.4 No water in excess of the quantity required in the approved mix shall be allowed to be added to the concrete to increase its workability affected by elapsed time and/or temperature. While it is not being discharged, the concrete shall be placed in its final position and left undisturbed within two hours from the time when the cement is added to the mix.

14.0 Concrete Mixes

14.1 The Contractor shall submit for the Engineer approval, details of the design mixes he proposed to use in the ready-mix concrete.

14.2 For each grade of concrete mix used, the Contractor shall state the weights of cement, granite and fine aggregates, the type and amount of admixtures and the water cement ratio to be used in the mix.

14.3 Concrete to be used for the installation of bored piles shall be of minimum Grade 40 for concreting in the dry. For tremie concreting under water or suspension / drilling fluid, the concrete shall be at least one grade higher than the design grade.

14.4 Notwithstanding the above, the slump for concrete as measured by the standards slump cone shall be between 100mm and 150mm and the cement content in any mix shall not be less than 350 kg/m3. Where concrete is to be placed underwater or where pile will be exposed to sea water, the cement content shall not be less than 400 kg/m3. For tremie concreting the slump for the concrete shall be between 150mm to 200mm.

14.5 Concrete to be used for ultimate load test piles shall have a minimum cube strength of 30 N/mm² at 7 days and before commencement of load test.

14.6 14.6 The Engineer reserves the right to instruct the Contractor to modify, alter and improve the concrete mix, if necessary.

15.0 Testing of Concrete

15.1 A minimum of 2 cubes shall be cast from every truck load of concrete for the purpose of compression tests.

15.2 The Engineer reserves the right to reject the pile and the Contractor shall install compensating piles that will ensure the safety of the Building to be supported by the piles. The compensating piles shall be installed at the Contractor's expense, and as directed by the Engineer.

15.3 The test cubes shall be made and cured in accordance with EN 12350, EN 12390, EN 12504 and BS 1881 or local code requirements. Concrete shall be placed in standard steel mould in layers approximately 50mm deep and each layer compacted by hand using a steel bar 300mm long weighing 1.8 kg and having a ramming face 25mm square. The number of strokes per layer shall not be less than 35 for 150mm cubes. The cubes shall be tested at an approved accredited laboratory at the Contractor's expense.

16.0 Concrete Strength

The compressive strength of the concrete shall be not less than the strength in the following table corresponding to the grade of concrete:

Concrete	Minimum Crushing Strength (N/mm²)	
	7 days	28 days
Grade C32/40	27	40
Grade C28/35	23	35
Grade C25/30	20	30

17.0 Steel Reinforcement and Stirrups

17.1 Unless otherwise specified, the minimum percentage of main reinforcement shall be 0.5% of the cross-sectional area of the pile.

17.2 The length of the reinforcement cage shall be as shown in the drawing, but no less than 10m long below the pile cut-off level or full length of pile if the pile is less than 10m long below cut cut-off level. The rebars cages to go beyond the Marine clay layer.

17.3 The lapping joint for the stirrups shall be 50mm (fully welded) or 200mm (without weld) and the weld of the stirrups to the main reinforcing bars shall be able to withstand all condition of handling, transportation and during concreting. The reinforcing cage shall not be distorted in any way during these processes and shall be vertical when lowered into the boreholes.

18.0 Testing of Steel Reinforcements

18.1 Steel reinforcement shall be tested by an approved accredited testing authority approved by the Engineer.

18.2 The reinforcement shall be tested in accordance with the relevant Singapore Standard. Certificates of origin and manufacturer’s test certificates shall be produced. Selection of test shall be made by the Engineer. The Contractor shall bear all cost involved in making these tests. The Contractor shall have no claim for bars mutilated in obtaining the test pieces.

18.3 The Contractor shall furnish, for reinforcement supplied by him, manufacturer’s certificates and these shall be submitted for approval by the Engineer before any materials is brought on to the site.

18.4 The Contractor shall also submit test record issued by accredited laboratory giving the test results of the mechanical properties including tensile strength, yield point and elongation.

19.0 Concrete Annular Spacers

Annular concrete spacers of 50mm thickness to provide 75mm cover to the main reinforcement shall be used. The spacers shall be made of Grade 25/30 concrete (Depending upon the pile concrete grade) and placed at not more than one meter below the top of the reinforcement cage and at not more than 4 meters interval measured along the main reinforcement.

20.0 Distribution of Concrete

The concrete shall be distributed from the mixer to the final position of placing b buckets or tremie pipes with bottom open doors or flaps or other approved means which do not caused segregation or otherwise impair the quality of the concrete. The maximum free fall of the concrete shall not be more than 2m or as directed by the Engineer.

21.0 Compaction

21.1 With a properly designed mix of suitable workability it shall not be necessary to vibrate the concrete, but attention shall be paid to the compaction of the concrete in the top 1.5m or so of the pile.

21.2 If in the opinion of the Engineer the concrete in the top shall be vibrated, the Contractor shall do so without extra cost.

21.3 In circumstances where the cut-off level is below the ground water level, the need to maintain pressure on the unset concrete, equal to or greater than the ground water pressure, is required.

22.0 Boring and Concreting of Piles

22.1 Boring shall be carried down to such soil strata determined from results of ground boring and soil test carried out previously as agreed by the Engineer.

22.2 The sides of all boring shall be kept intact and no loose material shall be permitted to fall to the bottom of the bored holes. The Contractors equipment shall be capable of sinking a steel casing to support and keep the side of the borehole intact when directed by the Engineer.

22.3 In sites where casings are not necessary to prevent collapse of sides of boreholes, temporary casings of at least one size bigger than the bored holes shall be placed around the boreholes before the inspection of boreholes by the Engineer or his representative and at least 1.2 meter above the ground to prevent fall of loose materials or persons into the boreholes. All boreholes waiting for concreting shall be set up with the temporary casing at all times.

22.4 The bases of all bored holes shall be thoroughly cleaned with cleaning buckets or other suitable means and inspected by the Engineer or his representative in site prior to concrete placing. All loose materials likely to affect the strength of the pile shall be removed forthwith by the Contractor to the satisfaction of the Engineer and at the Contractor's expense.

In cases where stabilising solution or water are used to prevent collapse of sides of boreholes, additional cleaning of bases shall be carried by air-lift before concrete is poured.

22.5 The Contractor must provide all facilities to enable the Engineer to carry out his inspection safely.

22.6 Holes reamed out at the base shall likewise be clean and firm before placing of concrete.

22.7 If ground water is found in the borehole and it affects the excavation and removal of soil from the borehole, or if it causes the collapse of the sides of the borehole, or if the side collapses irrespective of the presence of water, then steel casing of appropriate size and length shall be installed in the bored hole. In addition, approved chemical or polymer solution of sufficient strength to maintain the sides of the boreholes shall be utilized by the Contractor if needed.

Such steel casings if necessary, shall be supplied and installed by the Contractor without additional charges to the owner.

22.8 The steel casing must be sunk as deep as the borehole or the approved chemical or polymer solution must be maintained at a level within the borehole at least 500mm higher than the surrounding ground level during the whole of the boring operation.

22.9 All concrete placed in boreholes shall be lowered by tremie pipes with exceptional care so as to prevent segregation. Tremie pipes extending to full depth shall be used for concreting if ground water is present in the borehole

22.10 The top of the pile shall be brought up to at least 500mm above the cut-off level the pile to permit all laitance and weak concrete to be removed and to ensure that it can be properly keyed into the pile-cap. Any defective concrete in the head of the completed pile should be cut away and made good with new concrete well bonded into the old concrete.

22.11 All boreholes shall be protected from the possibility of ground surface water entering the hole from the time the boring is completed and ready to be concreted until the pile is completed. No concreting shall be commenced until the hole has been inspected and the above precautions made.

22.12 Reinforcement cages shall be supported centrally and firmly within the hole so that that no movement will occur during the concreting operation.

23.0 Earth from Borehole

Excavated materials from the boreholes and pile-caps shall be removed from the site to Contractor’s own dumping grounds and at his own cost.

24.0 Backfilling Above Pile Casting Level

The Contractor shall backfill the empty bore between the top of concrete pile and ground surface with sand or excavated materials approved by the Engineer.

25.0 Position and Alignment Tolerances

25.1 The centre line of each pile shall not deviate at any point from its true position, as indicated on the drawings by more than 75mm on plan in any direction.

25.2 The verticality of each pile shall not deviate at any point below the ground by more than 1 : 75.

26.0 As-built Piling Plans

26.1 Submission of As-Built Plans

(a) On completion of pile installation, the Contractor shall within 14 days after completion, submit the as-built piling plans (1 soft copy in latest version of AutoCad and 1 paper print) incorporating pile penetrations, and location and type of pile load tests carried out.

(b) The Contractor shall submit the as-built piling plans (1 soft copy in latest version of AutoCad and 1 paper print, certified by a Licensed Surveyor) incorporating records of pile eccentricities within 10 days after pile-tops have been exposed. For the survey measurement of eccentricities of piles with pile-tops installed below specified levels, it will be the responsibility of the piling Contractor to excavate and expose the piles to enable such measurements to be carried out. Such works shall be carried out without additional costs.

(c) For projects having more than one block of building (linked or otherwise), submission of the above as-built piling plans by the Contractor shall be on a block by block basis.

26.2 Progressive Submission of Pile Eccentricities of Pile Groups

Progressively upon exposure of pile heads by the Building Contractor, the piling Contractor shall immediately survey the exposed piles and submit the as-constructed pile location drawings indicated clearly, the eccentricities of each pile of every pile-group. Such surveys shall be carried out by a Licensed Surveyor.

27.0 Design of Remedial Works for Excessive Single-Pile / Pile Group Eccentricities

The piling Contractor shall be entirely responsible for design proposals of remedial works for single-piles and group-piles having eccentricities greater than 75mm. The design proposals and drawings of the remedial works shall be undertaken by a local Professional Engineer engaged by the piling Contractor. Such remedial works shall include the provisions of additional piles, pile-caps, tie-beams, stumps, and / or redesign of pile-cap tie-beams and stumps, as deemed necessary by the Consulting Engineer. Proposals submitted by the Contractor shall be subject to the approval of the Consulting Engineer.

Progressively upon the confirmation of the single-pile / group pile eccentricities by the piling Contractors registered surveyor, the Contractor shall carry out the design of the remedial works and submit his professional Engineer’s calculations and drawings to the Engineers within 4 days thereof for clearance.

28.0 Pile Cutting and Trimming

The piling Contractor shall provide and maintain sufficient workers and equipment to carry out the cutting / trimming and extension of piles to cut-off levels. There shall be sufficient resources to cut / trim and/or extend at least 10 numbers of installed bored piles within 2 days after receipt of a 24-hour notice from the main Contractor. The concrete rubble resulting from pile hacking shall be remove from site progressively so as not to cause obstruction to the substructure works.

29.0 Stripping Pile Head and Bonding

The Contractor shall strip the pile heads to cut-off level as required by the Engineer.

Main reinforcement of piles shall be exposed for bonding into the pile-cap. Minimum bond length of main reinforcement to be exposed shall be 40 times the diameter of rebar.

In stripping pile heads, the concrete shall be stripped to a level such that the remaining concrete will project 75mm into the pile-caps.

30.0 Bending Straight Steel Reinforcement

The Contractor shall during trimming of the pile-heads bend straighten all reinforcements protruding from cut-off level of piles.

31.0 Damaged or Rejected Piles

Where a pile had been damaged during / after installation, testing or by other causes, the damaged pile shall be considered and treated as a faulty pile and shall be replaced by additional piles as approved by the Engineer at the Contractors' expenses.

Any pile not truly constructed and installed in accordance with this Specification shall be rejected and replaced by the Contractor at his own expense even if piles of bigger capacity are required.

Any extra cost of pile caps, beams etc. arising out of this shall be borne by the Contractor.

32.0 Adjacent Structures and Buildings

32.1 The Contractor shall ensure that no damage or nuisance is caused to existing structures and buildings in the vicinity of the site.

32.2 Before commencing the works the Contractor shall undertake a comprehensive survey of any adjacent land, buildings, structures, services and the like which may be affected. Such a survey shall clearly show the existing condition, a record of defects, extent of cracks, presence of underground structures and the location of services.

32.3 Colour photographs, reports and records of the adjacent land, buildings, structures, services and the like shall be made and a minimum of 36 negatives and 5 sets prints in albums, dated and suitable captioned together with 5 sets of the reports and records shall be supplied to the Engineer prior to commencement of works. Where appropriate, the Contractor shall make available copies of the photographs, reports and records to owners of the adjacent land, buildings, structures and the like. The Contractor shall employ an approved Loss Adjustor to provide this service.

32.4 The Contractor shall provide 'tell-tales' gauges or other suitable approved means including all instruments and equipment for regular recording and checking of cracks or other defects of the adjacent land, buildings, structures, services and the like.

32.5 The Contractor shall be solely responsible for the safety and stability of any adjoining structures and buildings and where necessary allow in his tender for any shoring and strutting required. Such shoring shall be so positioned or altered and adapted from time to time so as to maintain adequate working space for construction operations.

33.0 Continuous Monitoring of Foundation Settlement

33.1 The Contractor shall submit to the Engineer detailed proposals of continuous monitoring of settlement of foundation of existing properties, and provide all personnel, equipment and facilities to monitor continuously on a regular basis as determined by the Engineer (in any case not exceeding fortnightly) to an accuracy of 1mm, the foundation settlement of the building, surrounding roads, drains, walkways and the like throughout the progress of the works.

33.2 Compliance with Land Transport Authority's (LTA) or local authority Requirements:

The Contractor shall comply with the LTA's or local authority requirements as follow:

Submission of a certified survey plan indicating levels of existing abutting walls, drains and walkways at 5m grid for the Authority's record prior to commencement of piling works. During the piling works, levels must be monitored regularly and proper records furnished to LTA or local authority for retention.

All settlement monitoring works shall be carried out by a Registered Surveyor approved by the Engineer.

34.0 Construction Induces Vibration

34.1 The Contractor shall carry out the piling work in such a manner to minimise vibration.

34.2 Low vibration equipment to install and extract temporary casings for bores shall be used.

34.3 The Contractor shall submit a Method Statement of the precautionary measures that will be taken prior to and during piling works to avoid causing damages to neighbouring properties due to vibrations induced from piling works.

34.4 If necessary, the Contractor shall reduce vibration during pile driving by providing cut-off trenches 1.0 metre to 1.5 metre deep along the periphery of the site and any other precautionary measures approved by the Engineer, all at the expense of the Contractor.

35.0 Removal of Obstruction

35.1 The tender price shall include and allow for excavation in any material to remove obstruction not exceeding 2.0 meter below ground level which prevent deriving or which interfere with the proper alignment of the piles, timbering, backfilling, restoration of ground surrounding the piles and any other works necessary to complete the operation to the satisfaction of the Engineer.

35.2 No claims for standing time for rigs as a consequence of obstruction of any kind will be allowed.

36.0 Schedule of Construction Test for Concrete Works

The Contractor shall allow and conduct all construction tests a prescribed in the Building Control Regulation 1989 or equivalent or local code requirements.

Such tests shall be carried out by an Accredited laboratory.

Tests to be carried out	Standards to be used	Minimum frequency of testing/ the numbers of testing required
Cement	SS EN 197-1	}
Aggregate	BS 812 for compliance with SS EN 12620	} To be obtained from } concrete Supplier
Water	EN 1008	}
Concrete Cubes	EN 12350, EN 12390, EN 12504 and BS 1881	1 sample of 2 cubes per truck load
Steel Reinforcement :		
a) Tensile test	SS 560	1 sample of 3 test-pieces
b) Bend test and	SS 561	per size per consignment
c) Rebend test		

37.0 Ground Level Survey

The Contractor shall carry out pre-commencement and post-construction spot level surveys of the site platform at 5 meter grids. The survey shall be carried out by a registered Surveyor and based on an agreed bench mark.

38.0 Document Control

38.1 Approval: GLOBAL_FAC_GFTT_APPR

38.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_ NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY

38.3 Notification:

All Micron Team Members at all Sites

38.4 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

38.5 Review

Biennially (two years)

39.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New Document	07/01/2022